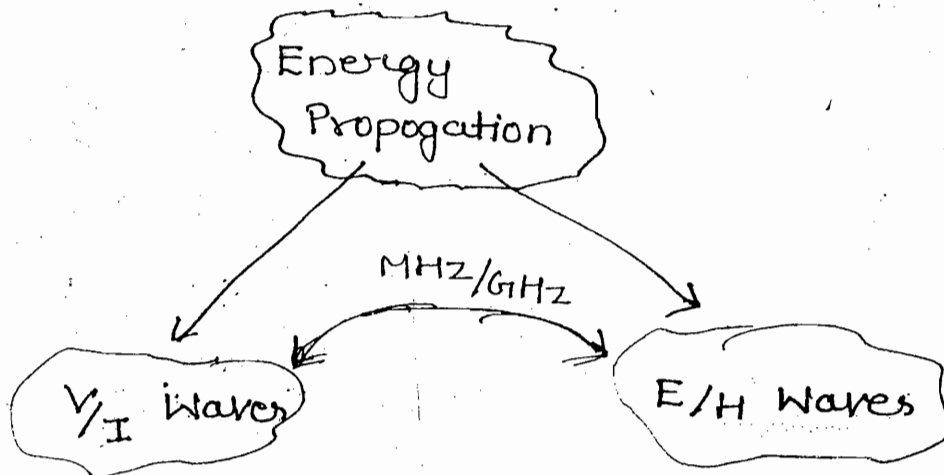
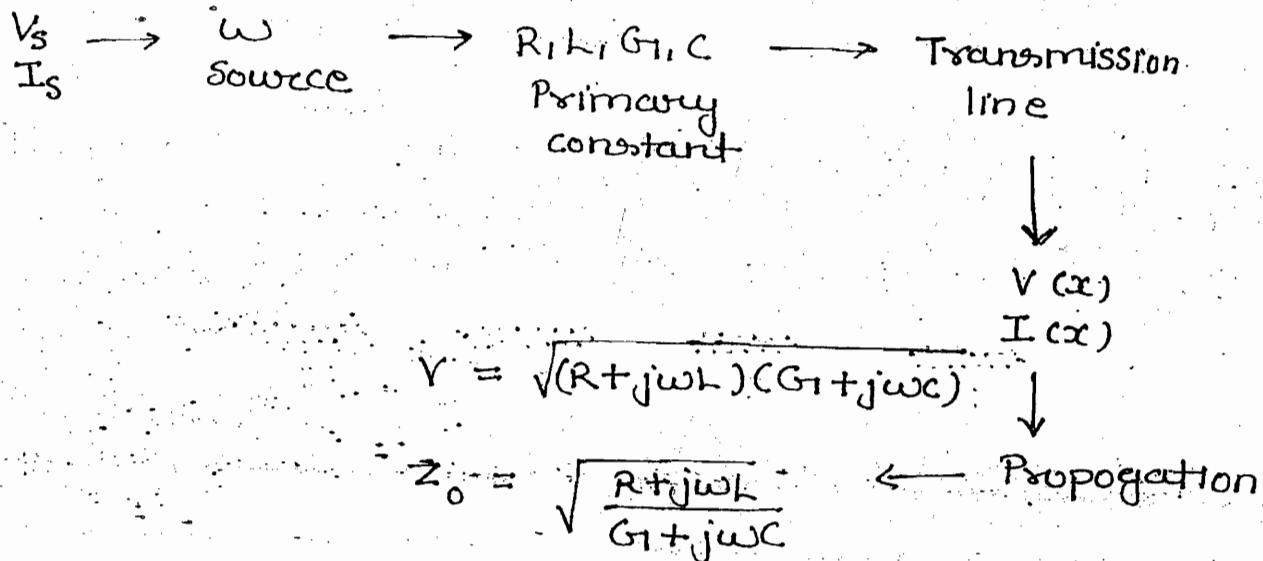


TRANSMISSION LINES



- $\epsilon = \infty$
- Conductors
- Wired or point to point
- Extremely low frequency
 $f < \text{kHz}$

- $\epsilon = 0$
- Free space
- Broadcast wireless
- Extremely high frequency
 $f > 10^{12} \text{ Hz}$



$E_s \rightarrow \omega$ source $\rightarrow \mu, \sigma, \epsilon$ Material constant $\rightarrow$ Medium $\rightarrow E(z)$
 $H_s \rightarrow H(z)$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

Primary constants:-

Resistance:-

→ It is the resistance of the conducting material all along the length of the line.

$$\rightarrow R = \rho \frac{l}{A} \quad R \cdot d \cdot l$$

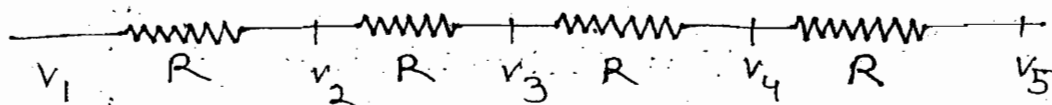
It is distributed all along the length

$$R = \frac{R}{l} = \text{ohm/m}$$

↓
Primary Constant

→ It is the resistance at any instance on the line

→ It appears in series along the line

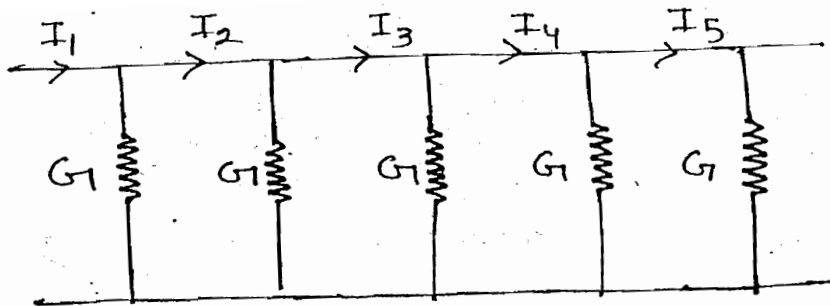


→ It causes a voltage decay all along the line

Conductance:-

→ It is the conductance of the insulating material b/w the lines

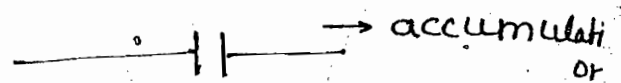
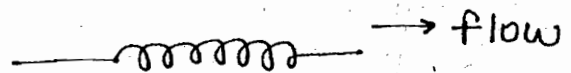
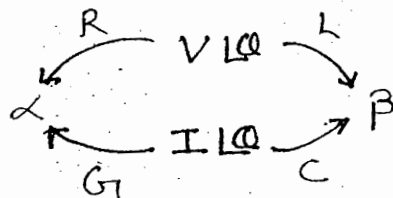
→ It appears in shunt b/w the lines



→ It causes current decay

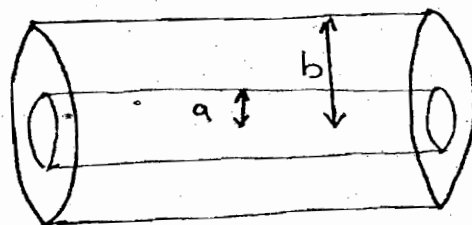
→ It is distributed all along the length

$$G_1 = \frac{G}{l} = \frac{\text{mho}}{\text{m}}$$

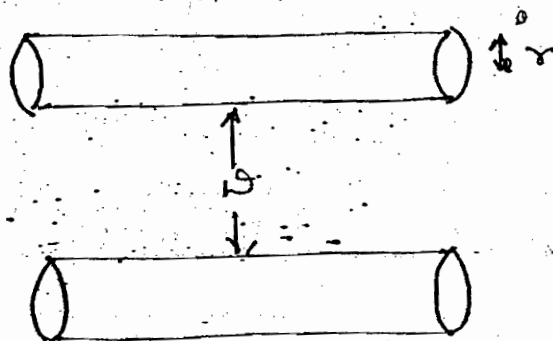


Capacitance:-

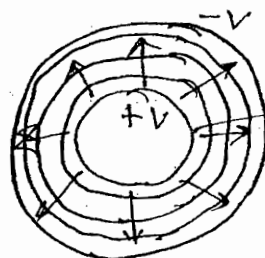
→ It is due to the voltages on the line there is a surrounded E field and hence capacitance



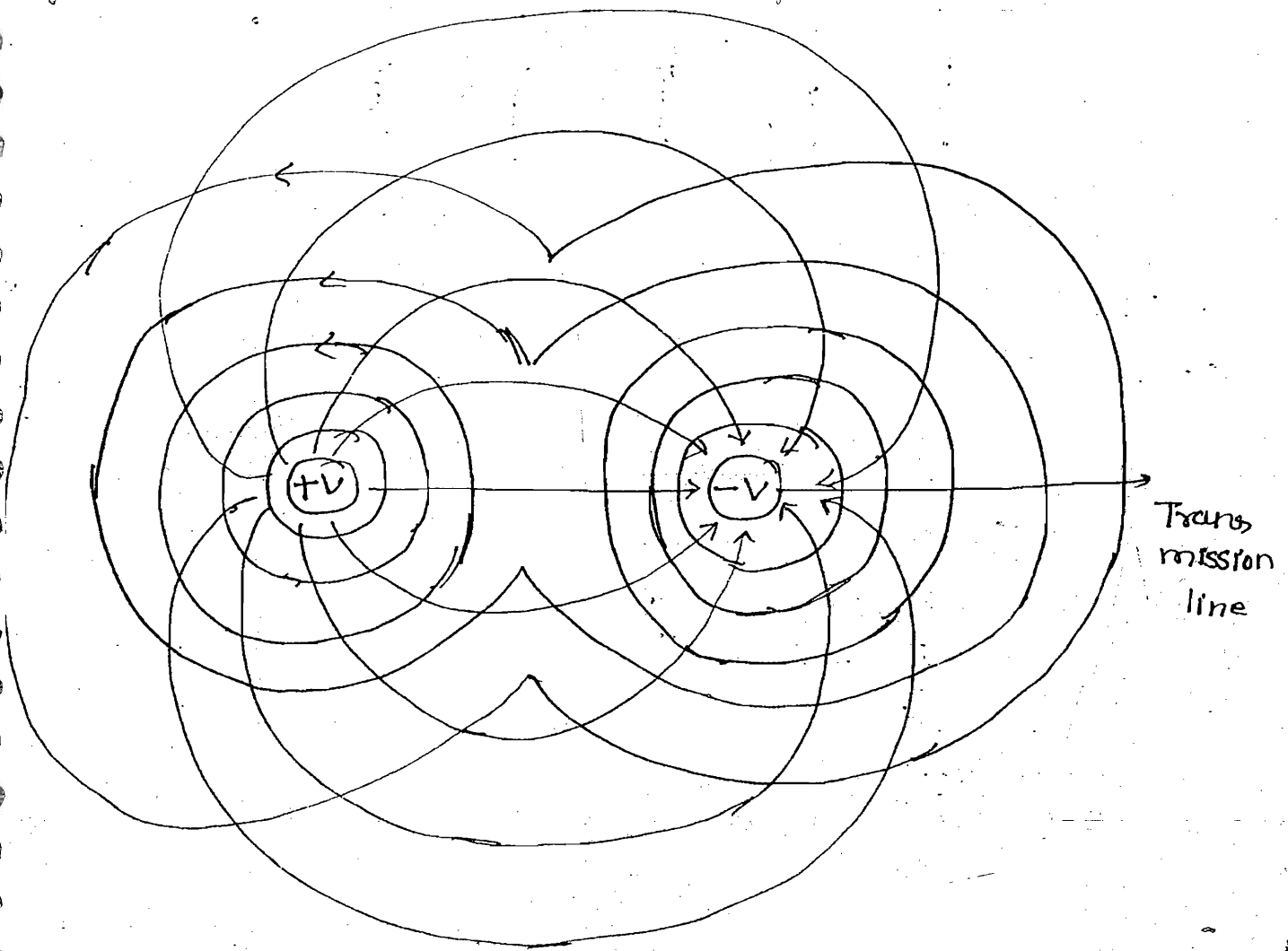
→ Co-axial cable



→ Parallel Wire.



→ Transmission line.



$$C = \frac{2\pi\epsilon_0 l}{\ln(b/a)} \rightarrow \text{Co-axial cable}$$

$$C = \frac{\pi\epsilon_0 l}{\ln(D/r)} \rightarrow \text{Parallel wire}$$

→ $C \propto l$ → capacitance is distributed

$$\rightarrow C = \frac{C}{l} = \frac{\text{Farads}}{\text{m}}$$

→ It appears in shunt b/w the cables

Inductance:-

→ It is due to the currents on the line there is a surrounded H-field and hence inductance

$$\rightarrow L = \frac{\mu l \ln(b/a)}{2\pi} \rightarrow \text{co-axial cable}$$

$$L = \frac{\mu l}{\pi} \cdot \ln(D/r) \rightarrow \text{parallel wire}$$

→ L is distributed i.e. $L \propto l$

$$L' = \frac{L}{l} \quad \frac{\text{Henry}}{\text{m}}$$

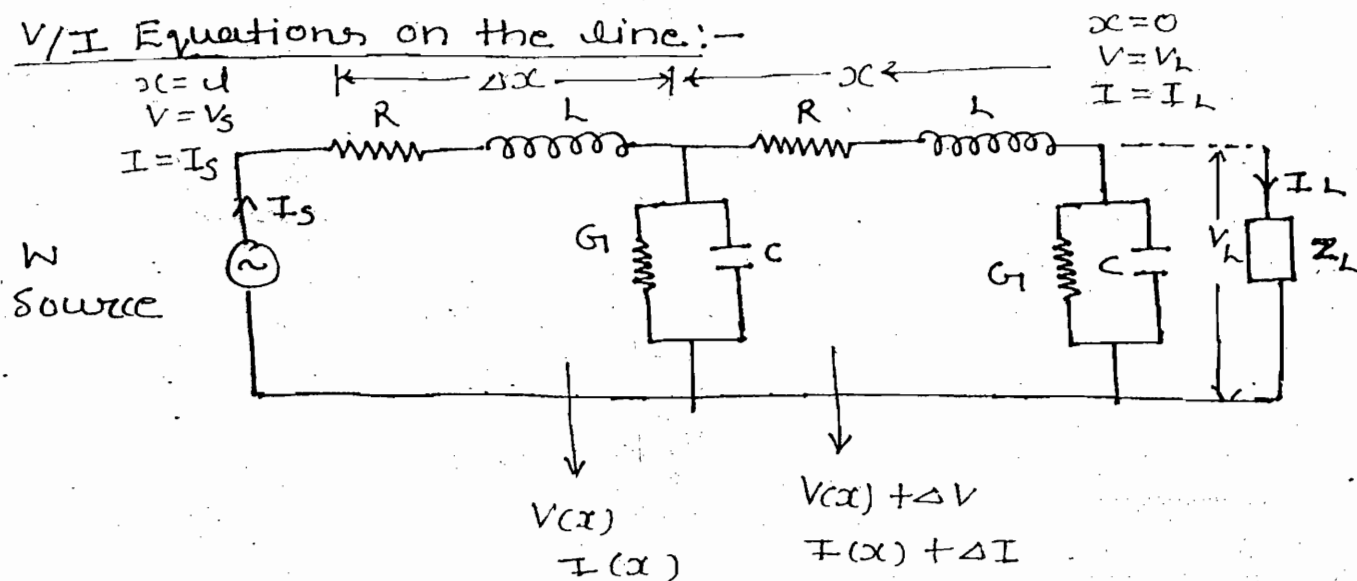
Note:-

$$LC = \text{Distributed } L \times \text{Distributed } C = \mu \epsilon$$

For any transmission line and for any geometry.

Lecture - 12

V/I Equations on the line:-



$$Z_{\text{series of } \Delta x \text{ length}} = (R + j\omega L) \cdot \Delta x$$

$$Y_{\text{shunt of } \Delta x \text{ length}} = (G + j\omega C) \cdot \Delta x$$

$$\Delta V = I (R + j\omega L) \cdot \Delta x$$

$$\frac{\Delta V}{\Delta x} = \frac{dV}{dx} = (R + j\omega L) I \quad \text{--- (I)}$$

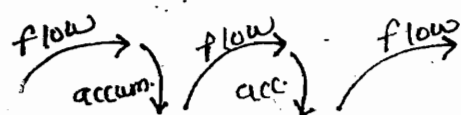
$$\frac{\Delta I}{\Delta x} = \frac{dI}{dx} = (G + j\omega C) V \quad \text{--- (II)}$$

Take I from (I) & put in (II)

$$\frac{d}{dx} \left[\left(\frac{1}{R + j\omega L} \right) \frac{dV}{dx} \right] = (G + j\omega C) V$$

$$\frac{d^2 V}{dx^2} = (R + j\omega L) (G + j\omega C) \quad \text{--- (III)}$$

$$\frac{d^2 I}{dx^2} = (R + j\omega L) (G + j\omega C) \quad \text{--- (IV)}$$



→ meaning of Harmonic

Note:-

If the source is time harmonic at one end the effect is space harmonic all along the length of the line.

Let $\sqrt{(R+j\omega L)(G+j\omega C)} = Y$ (per cm)
 Eq-(III) & (IV) can be written as Unit

$$\frac{d^2 V}{dx^2} - Y^2 V = 0$$

$$\frac{d^2 I}{dx^2} - Y^2 I = 0$$

The V/I solution on the line is

$$V(x) = c_1 e^{-Yx} + c_2 e^{Yx} \quad \text{--- (V)}$$

$$I(x) = c_3 e^{-Yx} + c_4 e^{Yx} \quad \text{--- (VI)}$$

Applying initial conditions, $(x=0, V=V_L, I=I_L)$

$$V_L = c_1 + c_2 \quad \text{--- (VII)}$$

$$I_L = c_3 + c_4 \quad \text{--- (VIII)}$$

Using eq-(I) in eq-(V)

$$\frac{dV}{dx} = -Y c_1 e^{-Yx} + Y c_2 e^{Yx} = (R+j\omega L) I$$

Put $x=0$

$$\begin{aligned} \frac{c_2 - c_1}{\sqrt{(R+j\omega L)(G+j\omega C)}} &= \frac{(R+j\omega L) I_L}{\sqrt{(R+j\omega L)(G+j\omega C)}} \\ &= \sqrt{\frac{R+j\omega L}{G+j\omega C}} I_L = I_L Z_0 \end{aligned}$$

$$\text{where } Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$I_L Z_0 = c_2 - c_1 \quad \text{--- (IX)}$$

$$\text{Similarly } \frac{V_L}{Z_0} = c_4 - c_3 \quad \text{--- (X)}$$

$$\boxed{c_1 = \frac{V_L - I_L Z_0}{2} \quad c_2 = \frac{V_L + I_L Z_0}{2}}$$

$$\zeta_3 = \frac{I_L - V_L/Z_0}{2}, \quad \zeta_4 = \frac{I_L + V_L/Z_0}{2}$$

$$V(x) = \frac{V_L}{2} \left[\left(1 - \frac{Z_0}{Z_L}\right) e^{-\gamma x} + \left(1 + \frac{Z_0}{Z_L}\right) e^{\gamma x} \right]$$

$$V(x) = \frac{V_L (Z_L + Z_0)}{2Z_L} \left[e^{\gamma x} + \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-\gamma x} \right]$$

$$V(x) = V_0 \left(e^{\gamma x} + \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-\gamma x} \right)$$

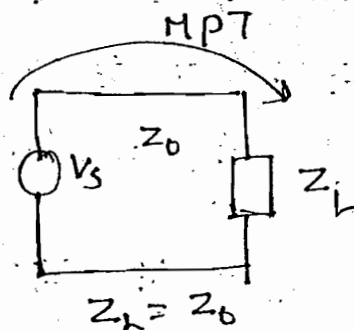
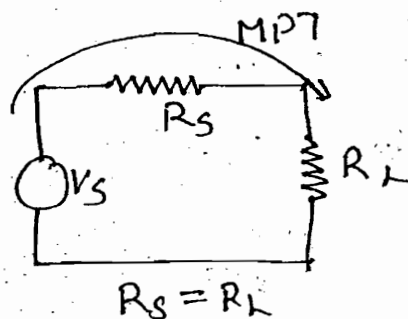
$$I(x) = \frac{I_L (Z_L + Z_0)}{2Z_0} \left[e^{\gamma x} + \left(\frac{Z_0 - Z_L}{Z_0 + Z_L} \right) e^{-\gamma x} \right]$$

$e^{\gamma x} \rightarrow -x$, Source-load forward wave

$e^{-\gamma x} \rightarrow +x$, load-source, reflected wave

→ Every transmission line has two waveforms i.e. the forward and reflected and the reflected wave does not exist in one single condition i.e. $Z_L = Z_0$. This is called as matched condition and this is a condition where complete source power is absorbed by the load.

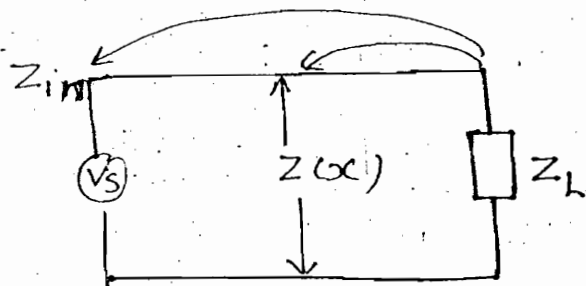
→ This is similar to max. power transfer in N/w or EM wave reflections b/w materials



Note:-

Z_0 is a unique impedance of the line with which if the line is terminated the reflections on the line and hence called as characteristic impedance

Impedance on line (Z) and i/p impedance (Z_{in}):-



Note:-

Due to the loaded one end and primary constant in the line, impedance is different on different point of line such that

$$Z(x) = \frac{V(x)}{I(x)} \quad Z_{in} = Z(l)$$

$$V(x) = \frac{V_L}{Z_L} \left(\frac{(Z_L + Z_0)}{2} e^{\gamma x} + \frac{(Z_L - Z_0)}{2} e^{-\gamma x} \right)$$

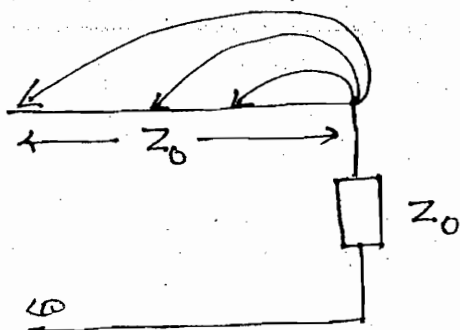
$$V(x) = V_L \cosh(\gamma x) + I_L Z_0 \sinh(\gamma x)$$

Similarly

$$I(x) = I_L \cosh(\gamma x) + \frac{V_L}{Z_0} \sinh(\gamma x)$$

$$Z(x) = Z_0 \left[\frac{Z_L \cosh(\gamma x) + Z_0 \sinh(\gamma x)}{Z_0 \cosh(\gamma x) + Z_L \sinh(\gamma x)} \right]$$

If $Z_L = Z_0$ then $Z(x) = Z_0 = Z_L = Z_{in}$



Matched line

→ For a matched line the impedance anywhere on the line is also same Z_0

Note:-

Z_0 is that unique impedance with which if the line is terminated impedance on the line is also the same

Short circuit and open circuit lines:-

→ If $Z_L = 0 \Rightarrow$ SC line

$$Z_{in} = Z_{sc} = Z_0 \tanh(\gamma x)$$

→ If $Z_L = \infty \Rightarrow$ O.C line

$$Z_{in} = Z_{oc} = Z_0 \coth(\gamma x)$$

$$\Rightarrow \boxed{Z_0 = \sqrt{Z_{sc} \cdot Z_{oc}}}$$

It is geometric mean of S.C and O.C i/p impedances.

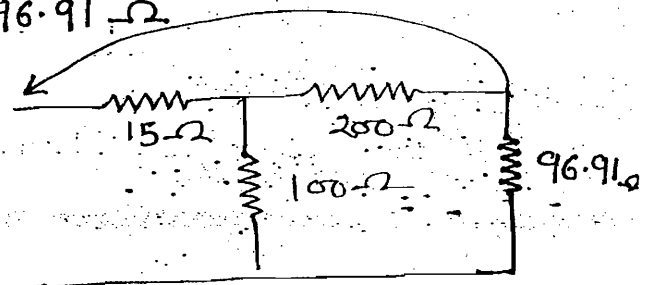
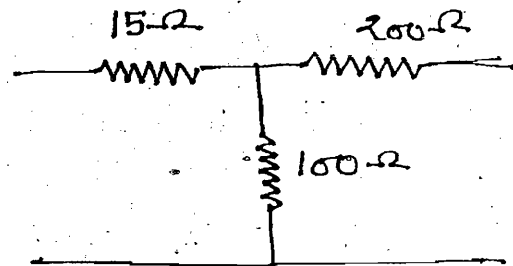
eg:-

$$Z_{oc} = 115 \Omega$$

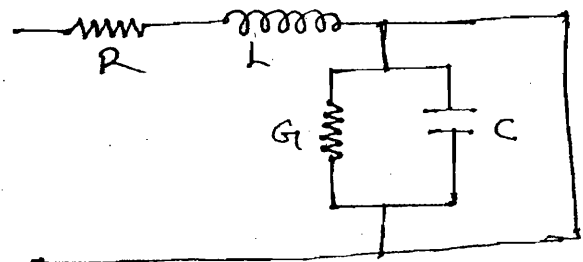
$$Z_{sc} = 15 + \frac{100 \cdot 200}{100 + 200}$$

$$= 81.67 \Omega$$

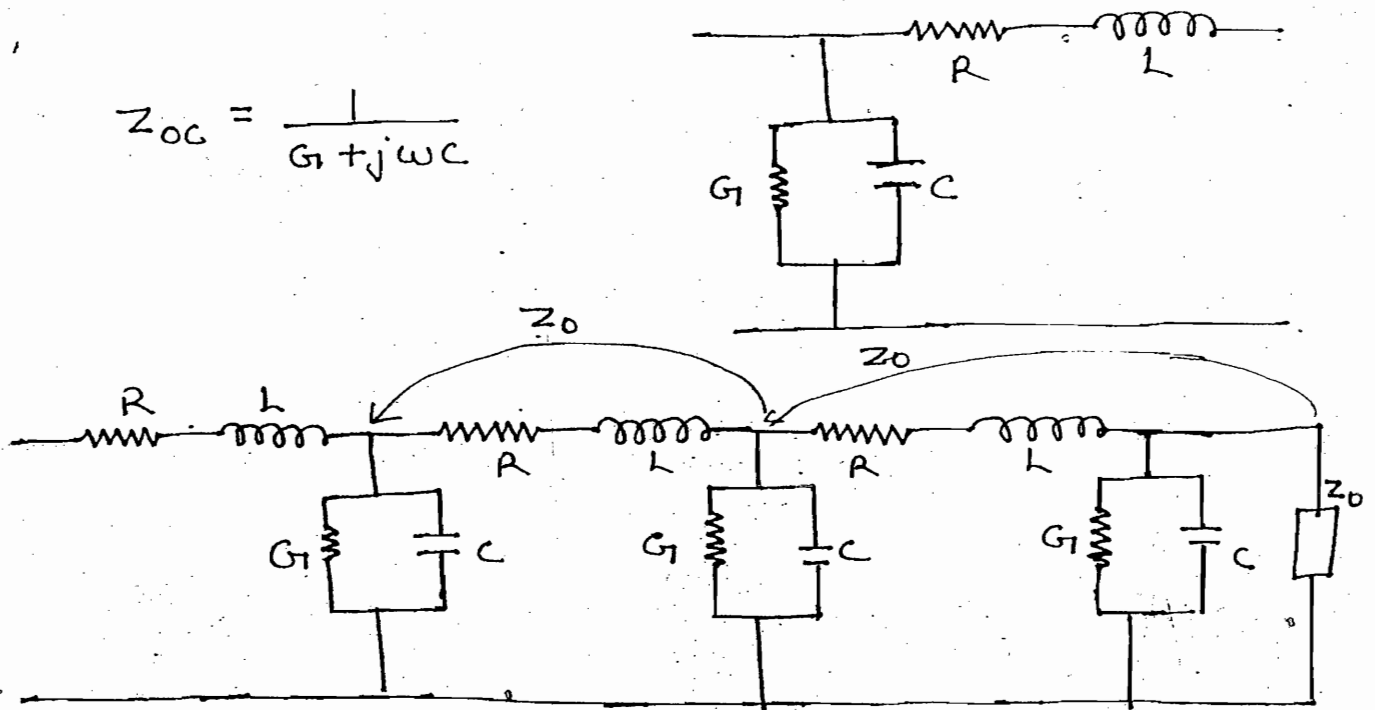
$$Z_0 = \sqrt{81.67 \times 115} = 96.91 \Omega$$



$$Z_{sc} = R + j\omega L$$



$$Z_{oc} = \frac{1}{G + j\omega C}$$



Note:-

- Any N/w either discrete or continuous when terminated with its characteristic impedance the same impedance appears on other end
- For a transmission line whose every RLC branch having Z_0 characteristic impedance when terminated with Z_0 at one end the loading effect appears to be the same anywhere on the line.
- Lossless line and distortionless line,

$$Y = \sqrt{R}$$

Lossless line and distortionless line:-

$$Y = \sqrt{(R + j\omega L)(G + j\omega C)} = \text{propagation constant} \\ = \alpha + j\beta$$

If $\alpha = 0$ everywhere on the line, the line is said to be lossless

$$(I) R = G = 0$$

$$(II) \left. \begin{array}{l} \omega L \gg R \\ \omega C \gg G \end{array} \right\} \text{High frequency lines}$$

$$Y = j\omega \sqrt{LC} \quad (\text{purely imaginary})$$

$$= j\beta$$

$$\beta = \omega \sqrt{LC}$$

$$v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\mu_0 \epsilon_0 \epsilon_R}} = \frac{3 \times 10^8}{\sqrt{\epsilon_R}}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{L}{C}} = \text{real}$$

$$= \sqrt{\frac{\mu_0 \ln(b/a) \ln(b/a)}{2\pi \cdot 2\pi \epsilon}}$$

$$= \sqrt{\frac{\mu_0}{\epsilon_0 \epsilon_R}} \cdot \frac{\ln(b/a)}{2\pi}$$

$$\boxed{Z_0 = \frac{60 \ln(b/a)}{\sqrt{\epsilon_R}}} \rightarrow \text{Co-Axial cable}$$

→ Z_0 is a design aspect of the line and depends on the cable dimensions but not on R, L, G, C or operational frequency of the line.

$$\boxed{Z_0 = \frac{120 \ln(b/a)}{\sqrt{\epsilon_R}}} \rightarrow \text{Parallel wire}$$

Distortionless line:-

If the phase is linear the wave is said to be distortionless i.e. $\beta \propto \omega \rightarrow$ For distortionless line

→ Equal Rise time = Full time is characteristic of distortionless line.

Series arm time constant = Shunt arm time constant

$$\frac{L}{R} = \frac{C}{G} \Rightarrow \boxed{LG = RC}$$

$$Y = \sqrt{R \left(1 + j\omega L/R\right) G \left(1 + j\omega C/G\right)}$$

$$= \sqrt{RG} \left(1 + \frac{j\omega L}{R}\right) = \text{Complex}$$

$$Z_0 = \sqrt{\frac{R \left(1 + j\omega L/R\right)}{G \left(1 + j\omega C/G\right)}} = \sqrt{\frac{R}{G}} = \text{Real}$$

Note:-

Every lossless line $\Rightarrow$ Distortionless $\nRightarrow$ lossless

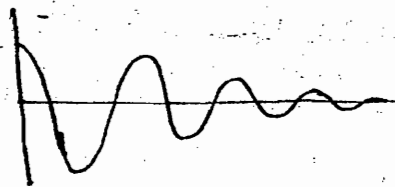
$$R = G = 0 \longrightarrow LG = RC$$

$$\beta = \omega \sqrt{LC} \longrightarrow \beta \propto \omega$$

Workbook:-

$$1) \quad Z_{in} = Z_0 \left[\frac{Z_L \cosh(j\beta l) + Z_0 \sinh(j\beta l)}{Z_0 \cosh(j\beta l) + Z_L \sinh(j\beta l)} \right]$$

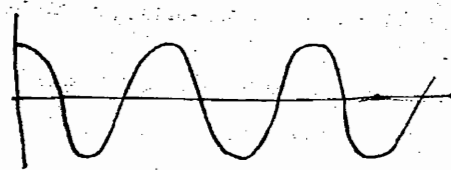
$$\cosh \theta = \frac{e^{\theta} + e^{-\theta}}{2}$$



$$\sinh(j\theta) = j \sin \theta$$

$$\cosh(j\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

↓ = $\cos \theta$



$$Z_{in} = Z_0 \left[\frac{Z_L \cos \beta l + j Z_0 \sin \beta l}{Z_0 \cos \beta l + j Z_L \sin \beta l} \right]$$

$$(i) \quad l = \frac{\lambda}{8} \quad \beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{8} = \frac{\pi}{4}$$

$$Z_{in} = Z_0 \left[\frac{Z_L + j Z_0}{Z_0 + j Z_L} \right]$$

$$(ii) \quad l = \lambda/4 \quad \beta l = \frac{\pi}{2} \quad Z_{in} = \frac{Z_0^2}{Z_L}$$

$$(iii) \quad l = \frac{\lambda}{2} \quad \beta l = \pi \quad Z_{in} = Z_L$$

$$Z_L = j50 \quad Z_0 = 50 \Omega$$

$$Z_{in} = 50 \left[\frac{j50 + j50}{50 + j50} \right] = \infty \rightarrow o/c$$

$$Z_{in} \quad (l/4) = \frac{50 \times 50}{j50} = -j50 \Omega$$

$$Z_{in} \quad (l/2) = Z_L = j50 \Omega$$

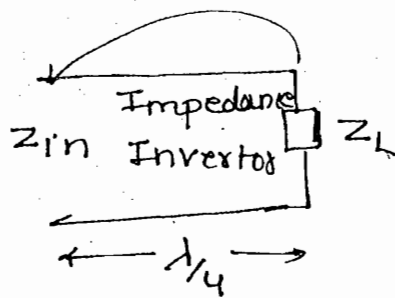
Note 1:-

→ βl is a critical aspect of the line that decides the nature and performance of the line and it is called an electrical path line.

→ length/ λ relationship is a critical aspect.

Note 2:-

$$l = \lambda/4 \quad Z_{in} = \frac{Z_0^2}{Z_L}$$



It's one end impedance has opposite nature to other end impedance hence called as impedance inverter

Note 3:-

$$l = \lambda/2 \quad Z_{in} = Z_L$$

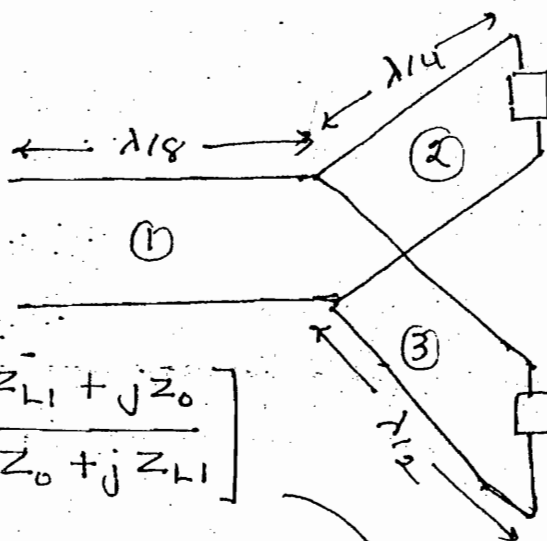
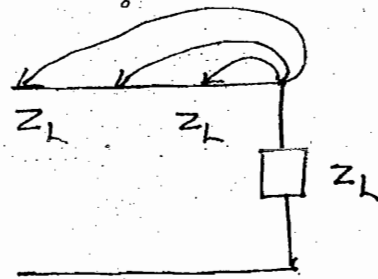
It is called as impedance reflector

Note 4:-

If $l \rightarrow 0$ or $l \rightarrow \infty$ where $\beta l \rightarrow 0$

$$Z(x) = Z_L$$

For electrically short line the wire is a simple connecting wire or a short ckt



$$Z_{in2} = \frac{R_0^2}{R_0/2} = 2R_0$$

$$Z_{in1} = Z_0 \left[\frac{Z_L + jZ_0}{Z_0 + jZ_L} \right]$$

$$Z_{in3} = 2R_0$$

$$Z_L = Z_{in2} \parallel Z_{in3} = R_0 \left[\frac{R_0 + jR_0}{R_0 + jR_0} \right]$$

$$= R_0$$

When $z_{in2} = \frac{R_0^2}{2R_0} = \frac{R_0}{2}$ Interchanged

$$z_{in3} = \frac{R_0}{2} \quad (\lambda/2)$$

$$z_{in1} = z_0 \left[\frac{z_{L1} + jz_0}{z_0 + jz_{L1}} \right] = R_0 \left[\frac{R_0/4 + jR_0}{R_0 + j\frac{R_0}{4}} \right]$$

$$= R_0 \left[\frac{1 + j4}{4 + j} \right]$$

$$Y_{in} = Y_{in1} + Y_{in2}$$

$$Y_{in1} = \frac{z_L}{z_0^2} = 0 \quad (\lambda/4)$$

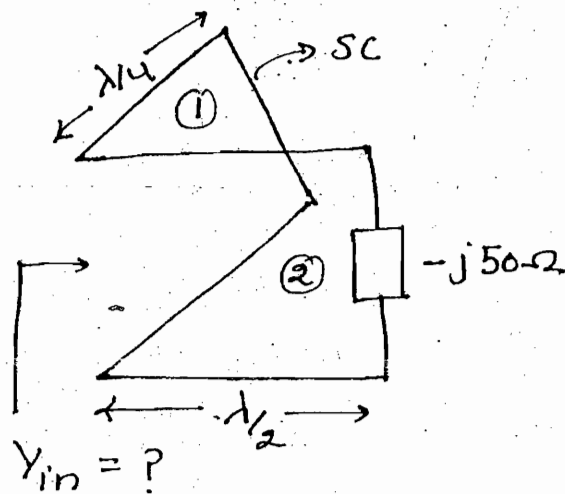
$$Y_{in2} = \frac{1}{z_L} = \frac{1}{-j50} \quad (\lambda/2)$$

$$= j0.02$$

$$Z_{in} = (Z_{in1} \parallel Z_{in2})$$

$$= (\infty \parallel -j50) = -j50$$

$$Y_{in} = j0.02 \Omega$$



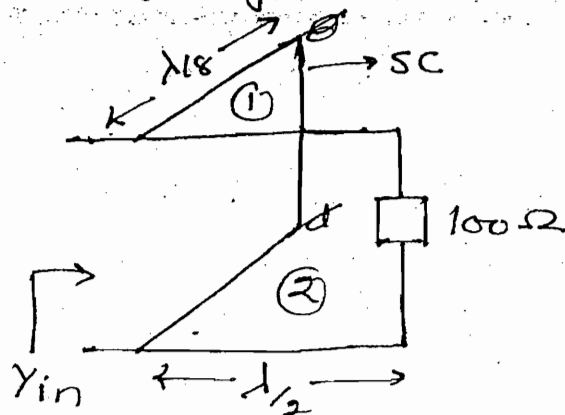
Stub $\Rightarrow$ SC line of finite length

$$Y_{in} = Y_{in1} + Y_{in2}$$

$$z_{in1} = z_0 \left[\frac{0 + jz_0}{z_0 + j0} \right] \quad (\lambda/8)$$

$$= jz_0 = j50 \Omega$$

$$Y_{in1} = \frac{1}{j50} = -j0.02$$



$$Y_{in2} = \frac{1}{Z_L} = \frac{1}{100} = 0.01$$

$$Y_{in} = Y_{in1} + Y_{in2} = 0.01 - j0.02 \text{ S}$$

8.

$$l = \lambda/4$$

$$Z_{in1} = R_0 \left[\frac{0 + jR_0}{R_0 + j0} \right]_{SC}$$

$$= jR_0$$

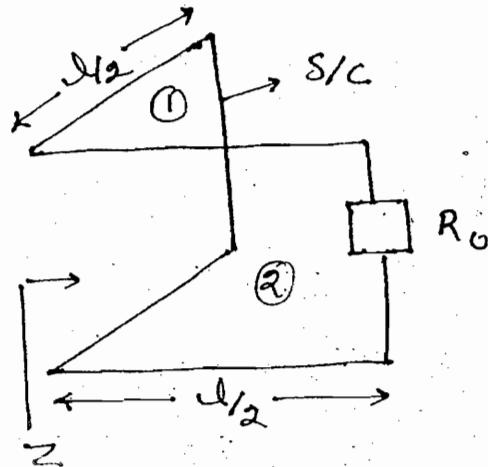
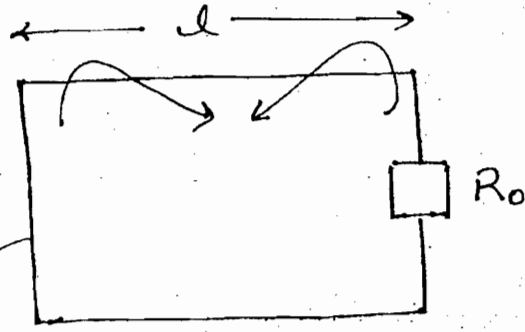
$$Z_{in2} = R_0$$

$$Z = \frac{jR_0 \cdot R_0}{jR_0 + R_0}$$

$$= \frac{jR_0}{j+1}$$

$$l = \lambda \quad Z = R_0$$

$$l = \lambda/8 \quad Z = SC$$



9.

$$V_p = 2 \times 10^8$$

$$f = 10 \text{ MHz}$$

$$\lambda = 20 \text{ m}$$

$$l = \lambda/2$$

$$l = 10 \text{ m}$$

$$Z_{in} = Z_L = 30 - j40 \Omega$$

10.

$$f' = 5 \text{ MHz}$$

$$\lambda' = 40 \text{ m}$$

$$l = \frac{\lambda'}{4}$$

$$Z_{in} = \frac{Z_0^2}{Z_L} = \frac{50 \times 50}{30 - j40} = 30 + j40$$

11.

$$Z_{in} = j60 = Z_{sc} = Z_0 \tanh(j\beta l)$$

$$= jZ_0 \tan \beta l$$

$$\beta l = \frac{\pi}{4}$$

$$= jZ_0$$

$$Z_0 = 60 \Omega$$

$$f' = 2f$$

$$12 \text{ MHz} \rightarrow 24 \text{ MHz}$$

$$\lambda' = \lambda/2$$

$$\beta l = 2\beta l = \pi/2$$

$$Z_{in} = Z_{sc} = jZ_0 \tan \pi/2 = \infty$$

$$f = 12 \text{ MHz}$$

$$Z_{in} = j60 = j\omega L \Rightarrow L = \frac{60}{2\pi \times 12 \times 10^6} = \frac{2.5}{\pi} \mu\text{H}$$

Summary:-

$$Z_{sc} = jZ_0 \tan \beta l$$

$$\left. \begin{array}{l} 0 < \beta l < \frac{\pi}{2} \\ 0 < l < \lambda/4 \end{array} \right\} Z_{in} \text{ is inductive}$$

$$\left. \begin{array}{l} \frac{\pi}{2} < \beta l < \pi \\ \lambda/4 < l < \lambda/2 \end{array} \right\} Z_{in} \text{ is capacitive}$$

$$Z_{oc} = -jZ_0 / \tan \beta l$$

$$Z_{in} \text{ inductive} \rightarrow Z_{in} \text{ is capacitive}$$

$$Z_{in} \text{ is capacitive} \rightarrow Z_{in} \text{ is inductive}$$

→ One end of a line is S.C or O.C then the other end is purely reactive

14. (a) $Z_{in} = -jZ_0$ For s/c line $\lambda/8$ length X
 $\downarrow$
 jZ_0

✓ (b) $Z_{in} = \pm j\infty$ for SC line $\lambda/4$ length

$$Z_{sc} = jZ_0 \tan \beta l$$

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

✓ (c) $Z_{in} = -jZ_0 \rightarrow$ o/c line $\rightarrow \lambda/2$ length



$$Z_{in} = Z_L$$

✓ (d) $Z_{in} = Z_0$

15. $Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{LC}{C^2}} = \frac{\sqrt{\mu_0 \epsilon_0 \epsilon_R}}{C} = \frac{\sqrt{\epsilon_R}}{v_p \cdot C}$

16. (a) $\alpha = \text{even}$, $\beta = \text{odd}$ X

(b) $\alpha = \text{constant}$, $\beta = k\omega$ ✓

(c) X $\left(\frac{L}{C} = \text{constant} = \frac{R}{G} \right)$

(d) Matched line ($Z_L = Z_0$) X

17. $\beta l = \omega \sqrt{LC} l = 108^\circ$

Reflection Coefficient:-

$$V(x) = \underbrace{V_0 e^{j\beta x}}_{V_f} + \underbrace{V_0 \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-j\beta x}}_{V_r}$$

$$\frac{V_r}{V_f} = \text{Reflection coefficient for the voltages anywhere on the line} = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-j2\beta x}$$
$$= \Gamma(x)$$

$$\boxed{\Gamma(x) = \frac{Z(x) - Z_0}{Z(x) + Z_0}}$$

$$\text{At } x=0 \quad \Gamma_{\text{at load}} = \frac{Z_L - Z_0}{Z_L + Z_0} = \text{complex}$$

$$= |\Gamma| \angle \theta = \left| \frac{V_r}{V_f} \right| \angle \theta$$

$|\Gamma|$ $\rightarrow$ It is the ratio of reflected voltage to the forward voltage amplitude

$\angle \theta$ $\rightarrow$ It is the time delay or phase difference b/w the reflected and forward voltage.

Γ is the measure of mismatch b/w the expected impedance and the actual impedance Z_L at the load.

$\Gamma(x)$ is the measure of mismatch b/w the expected impedance Z_0 and the actual impedance $Z(x)$ anywhere on the line.

Note:-

$$(i) \quad \Gamma_I = \frac{Z_0 - Z_L}{Z_0 + Z_L} = -\Gamma_V$$

If V_f and I_f they are in phase

$$V_0 e^{j\beta x} + V_0 \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-j\beta x}$$

$$I_0 e^{j\beta x} - I_0 \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-j\beta x}$$

then V_r and I_r are out of phase by 180°

(ii)

$$0 \leq |\Gamma| \leq 1$$

Matched line Complete Reflection (Mismatch)

Lecture - 13

Four Cases of Complete mismatch on a line ($|\Gamma|=1$):-

Case-(1):-

$Z_L = jR_0 = \text{Inductive load}$

$Z_0 = R_0 = \text{lossless line}$

$$|\Gamma| = 1$$

$$V_f \rightarrow \sin$$

$$V_r \rightarrow \cos$$

$$\Gamma = \frac{jR_0 - R_0}{jR_0 + R_0} = \frac{j-1}{j+1}$$

$$= \frac{\sqrt{2} \angle 135^\circ}{\sqrt{2} \angle 45^\circ}$$

$$= 1 \angle 90^\circ = j$$

Case-(II):-

$$Z_L = -jR_0 = \text{capacitive load}$$

$$Z_0 = R_0 = \text{lossless line}$$

$$\Gamma = \frac{-jR_0 - R_0}{-jR_0 + R_0} = \frac{-j-1}{-j+1}$$

$$= \frac{\sqrt{2} \angle 45^\circ}{\sqrt{2} \angle 135^\circ} = -j$$

Note:-

Real Power dissipation in any resistive load is always zero.

Case-(III):-

$$Z_L = 0 \Rightarrow \text{S.C load}$$

$$Z_0 = R_0 = \text{lossless line}$$

$$\Gamma = \frac{0 - Z_0}{0 + Z_0} = -1 = 1 \angle 180^\circ$$

$$V_f + V_r = 0$$

$$V_f \rightarrow \sin$$

$$V_r = -\sin$$

Voltage at S.C is always zero

Case-(IV):-

$$Z_L = \infty \Rightarrow \text{O.C load}$$

$$Z_0 = R_0 = \text{lossless line}$$

$$\Gamma = \frac{1 - Z_0/Z_L}{1 + Z_0/Z_L} = 1 \angle 0$$

$$V_f + V_r = V_{\max}$$

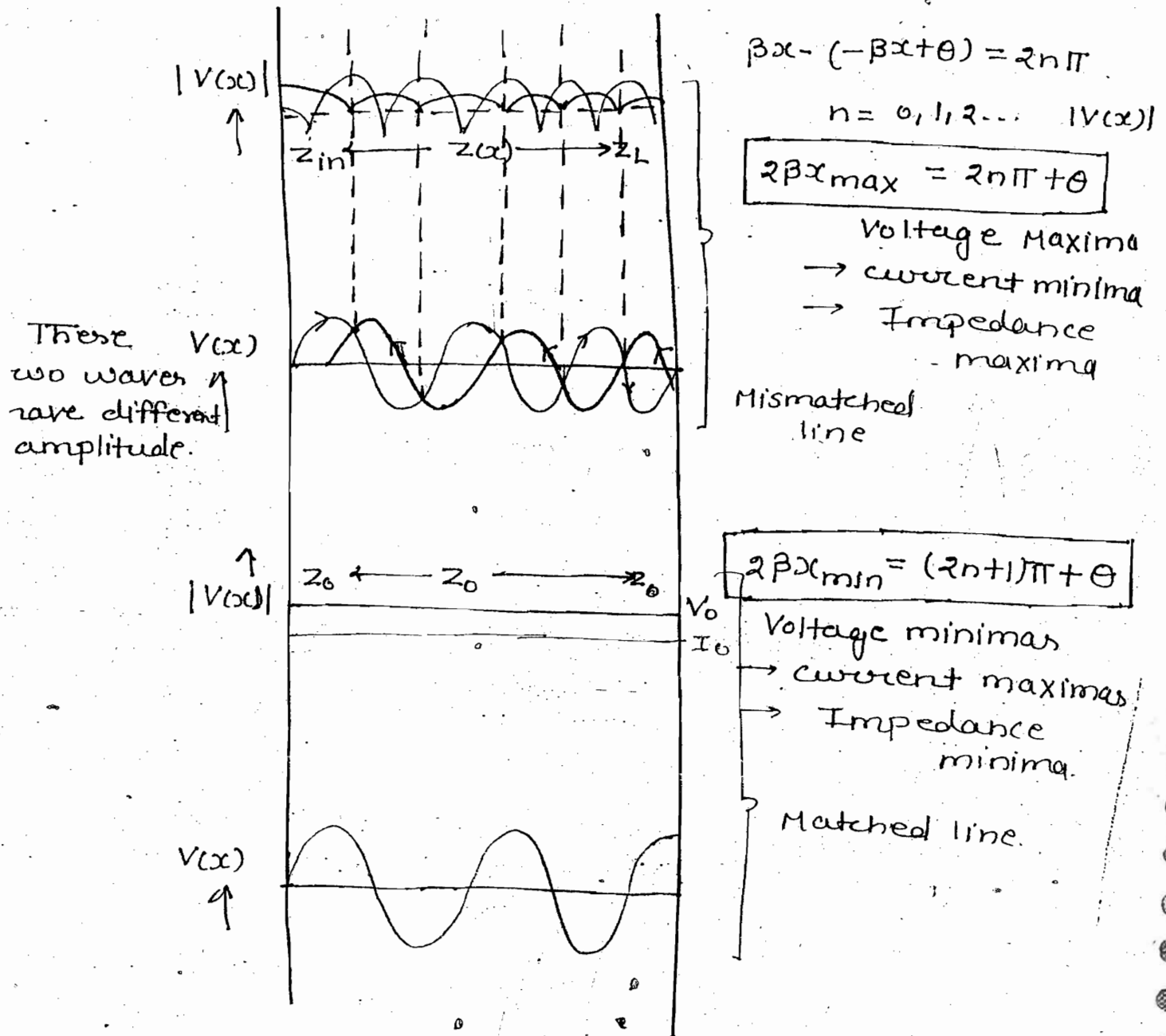
Voltage at O.C is always maximum

$$|\Gamma| = 1$$

$$V_f \rightarrow \sin$$

$$V_r = \sin$$

Standing Wave and SWR:-



$$V(x) = V_0 e^{j\beta x} + V_0 |\Gamma| e^{j\theta} e^{-j\beta x}$$

$$= V_0 e^{j\beta x} + V_0 |\Gamma| e^{j(-\beta x + \theta)}$$

The magnitude plot of a single harmonic matched line as a constant value straight line graph. The magnitude plot of two harmonics mismatched line as shown below:-

$$|V(x)| = V_{max} = V_0 + V_0 |\Gamma| \text{ when they are inphase}$$

$$\beta x - (-\beta x + \theta) = 2n\pi$$

$$|V(x)|$$

$n = 0, 1, 2$

$$2\beta x_{\max} = 2n\pi + \theta$$

Voltage Maxima Positions

$$|V(x)| = V_{\min} = V_0 - V_0 |\Gamma| \text{ when they are out of phase}$$

$$2\beta x_{\min} = (2n+1)\pi + \theta$$

→ Voltage Minima

→ The magnitude plot of two harmonics travelling in opposite direction having periodic interference results in addition and cancellation leading to standing wave formation

Note 1:-

The current also has identical standing wave pattern with the voltage maxima coincide with current minima

Note 2:-

As the impedance is different at different point on line voltage and current distribution is also extremely spread on the line such that

$$Z_{\max} = \frac{V_{\max}}{I_{\min}}$$

$$Z_{\min} = \frac{V_{\min}}{I_{\max}}$$

Note 3:-

$$\left. \begin{aligned} \beta x - \frac{2\pi}{\lambda} x &\rightarrow \text{Periodic} - 2\pi \\ x &\rightarrow \text{Periodic} - \lambda \end{aligned} \right\} \text{ Harmonic}$$

$$\left. \begin{aligned} 2\beta x - \frac{2\pi}{\lambda/2} x &\rightarrow \text{Periodic} - 2\pi \\ x &\rightarrow \text{Periodic} - \lambda/2 \end{aligned} \right\} \text{ Standing wave}$$

Maxima to Maxima distance on standing wave is $\lambda/2$

Standing Wave Ratio (SWR):-

It can be VSWR or CSWR

$$= \frac{V_{\max}}{V_{\min}} \text{ or } \frac{I_{\max}}{I_{\min}}$$

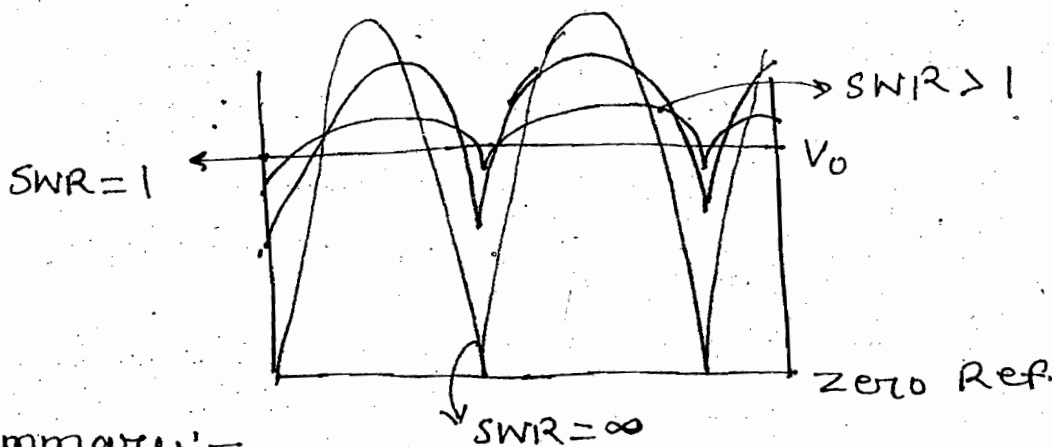
$$\boxed{\text{SWR} = \frac{V_0 + V_0 |\Gamma|}{V_0 - V_0 |\Gamma|} = \frac{1 + |\Gamma|}{1 - |\Gamma|}}$$

$$0 \leq |\Gamma| \leq 1$$

$$\boxed{1 \leq \text{SWR} < \infty}$$

Matched
line

Complete Mismatch



Summary:-

$$\left. \begin{array}{l} Z_0 \downarrow \uparrow \\ V_0 \uparrow \downarrow \\ I_0 \uparrow \downarrow \end{array} \right\} \text{SWR}$$

$$\begin{aligned} Z_{\max} &= \frac{V_{\max}}{I_{\min}} \\ &= \frac{V_0 + V_0 |\Gamma|}{I_0 - I_0 |\Gamma|} \end{aligned}$$

SWR is a measure of mismatch on the line

$$Z_{\max} = Z_0 \cdot \text{SWR}$$

$$Z_{\min} = Z_0 / \text{SWR}$$

$$ESWR = 5 = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$|\Gamma| = \frac{5-1}{5+1} = \frac{2}{3}$$

$$\Gamma = \frac{n_2 - n_1}{n_2 + n_1} = \frac{2}{3} \angle \theta = \pm \frac{2}{3}$$

$$n_2 + n_1 \rightarrow 120\pi \text{ (real)}$$

$$\frac{120\pi}{\sqrt{\epsilon_r}} \text{ (real)}$$

$$\text{As } n_2 < n_1$$

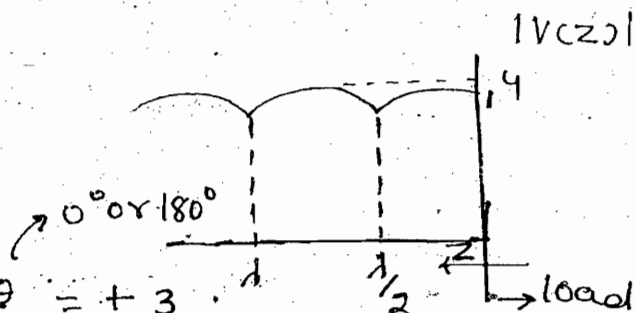
$$\Gamma = -\frac{2}{3} = \frac{n_2 - 120\pi}{n_2 + 120\pi} \Rightarrow n_2 = 24\pi$$

$$VSWR = 4$$

$$|\Gamma| = \frac{4-1}{4+1} = \frac{3}{5}$$

$$\Gamma = \frac{Z_L - 50}{Z_L + 50} = \frac{3}{5} \angle \theta = \pm \frac{3}{5}$$

$$\text{real} \rightarrow \text{real}$$



V_{min} is exactly at the load, Z_{min} is at the load

$$Z_L < Z_0$$

$\Rightarrow \Gamma$ is negative

$$\Gamma = -\frac{3}{5} = \frac{Z_L - 50}{Z_L + 50} \Rightarrow Z_L = 12.5\Omega$$

1st voltage maxima occurs at $\lambda/4$

$$2 \cdot \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = 2n\pi + \theta$$

$$\Rightarrow \theta = \pi$$

1 Note:-

2 $\rightarrow$ If Z_L, Z_0 are real
3 $Z_L > Z_0$

4 Γ is real & positive

5 Γ 's phase is 0

6 $2\beta x_{\max} = 2n\pi + 0 \Rightarrow x_{\max} = 0$

7 First voltage maxima exactly occurs

8 eg:- OC line where $\Gamma = +1$

9 $\rightarrow$ If Z_L, Z_0 are real

10 $Z_L < Z_0$

11 Γ is real and negative

12 First voltage minima exactly occurs

13 eg:- SC line where $\Gamma = -1$

14 $\rightarrow$

15
$$VSWR = \frac{1 + \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right|}{1 - \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right|}$$

16 If Z_L, Z_0 are real & $Z_L > Z_0$

17
$$VSWR = \frac{1 + \frac{Z_L - Z_0}{Z_L + Z_0}}{1 - \frac{Z_L - Z_0}{Z_L + Z_0}}$$

18
$$VSWR = \frac{Z_L}{Z_0}$$

19 If Z_L, Z_0 are real

20 $Z_L < Z_0$

21 $VSWR = ?$

24.

$$\Gamma(\lambda/8) = ?$$

$$\Gamma \text{ at load} = 0.6 \angle 60^\circ$$

$$\Gamma(x) = \frac{Z(x) - Z_0}{Z(x) + Z_0} = \frac{Z_L - Z_0}{Z_L + Z_0} e^{-j2\beta x}$$

$$\Gamma(\lambda/8) = \Gamma(0) e^{-j2\beta \cdot \frac{\lambda}{8}}$$

$$= 0.6 e^{j60} e^{-j2 \cdot \frac{2\pi}{\lambda} \cdot \frac{\lambda}{8}}$$

$$= 0.6 e^{j60} e^{-j90}$$

$$= 0.6 \angle -30^\circ$$

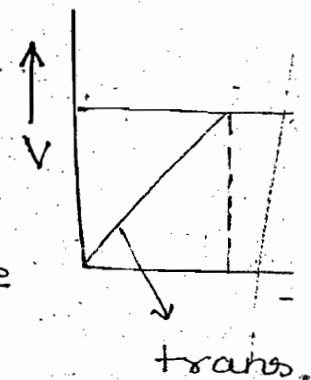
18+
← 17
7cm
h=0

25.

30V D.C Battery

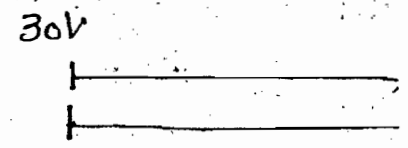
D.C Transients

During the transient state even DC has time variations and all AC fundamentals are applicable



$$V_f + V_r = \text{Adding}$$

$$\Gamma = \frac{10}{30} = \frac{Z_L - 50}{Z_L + 50}$$



$$\Rightarrow Z_L = 100 \Omega$$

$$V_f =$$

$$V_r$$

$$I_L = \frac{30}{Z_L}$$

(Steady state)

$$= \frac{30}{100}$$

$$= 0.3 \text{ Amp.}$$

n=0

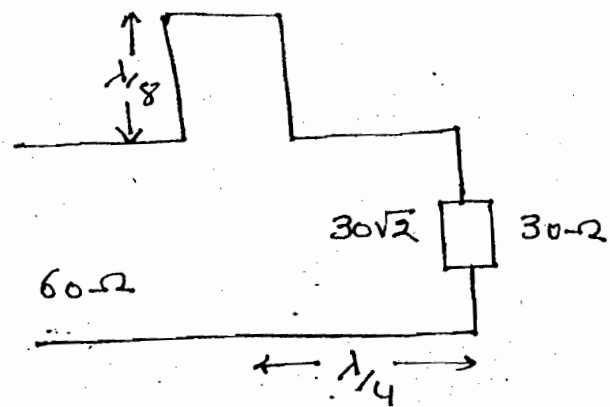
2.2

26.

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$= \frac{\sqrt{17} + 1}{\sqrt{17} - 1}$$

$$= 1.63$$



$$Z_{L1} = Z_{in2} + Z_{in3}$$

$$= 60 + j30 \Omega$$

$$Z_{in2} = j30 \Omega$$

$$Z_{in3} = \frac{(30\sqrt{2})^2}{30} = 60 \Omega$$

$$\Gamma = \frac{Z_{L1} - jZ_0}{Z_{L1} + jZ_0} = \frac{60 + j30 - 60}{60 + j80} = \frac{j}{j+4}$$

$$|\Gamma| = \frac{1}{\sqrt{17}}$$

27.

$$V_f = 25 \sin(\omega x - 75^\circ)$$

$$\Gamma = 0.6 \angle -30^\circ$$

$$V_r = ?$$

$$\frac{|V_r|}{|V_f|} = 0.6 \quad |V_r| = 15$$

$$V_r \text{ phase} = V_f \text{ phase} + \Gamma \text{ phase}$$

$$= -75^\circ - 30 = -105^\circ$$

$$V_r = 15 \sin(-\omega x - 105^\circ)$$

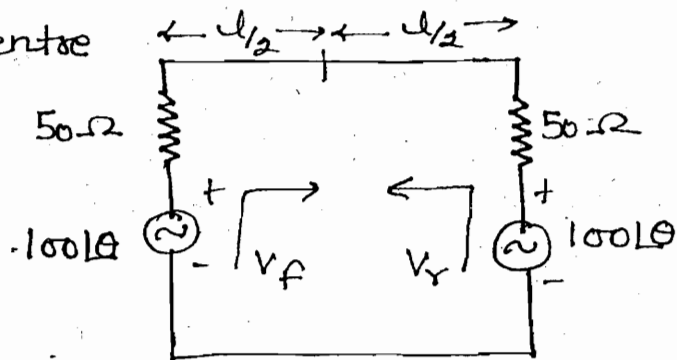
28. V_f & V_r are inphase
and add up at centre

I_f & I_r cancel
at the centre

$$|I_f| = |I_r|$$

$$\Rightarrow I_{\min} = 0$$

$$Z_{\max} = \frac{V_{\max}}{I_{\min}} = \infty$$



29.

$$VSWR = \frac{Z_L}{Z_0} = \frac{75}{50} = 1.5$$

Smith Chart (Circle Diagrams):

→ It is used to calculate Γ and VSWR for a known Z_L & Z_0 .

$$\begin{aligned} \Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} \\ &= \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} \end{aligned}$$

→ It is used normalised or relative ratio of Z_L/Z_0 for its calculation

→ It is a rectangular graph or polar plot of Γ_r vs Γ_i having two families of circles

→ constant R circles

where $R = \text{Real} [Z_L/Z_0]$

→ constant X circles.

$X = \text{Imag} [Z_L/Z_0]$

$$\Gamma_Y + j \Gamma_I = \frac{R + jX - 1}{R + jX + 1} = \frac{(R-1) + jX}{(R+1) + jX}$$

$$\left(\Gamma_Y - \frac{R}{R+1} \right)^2 + \Gamma_I^2 = \left(\frac{1}{R+1} \right)^2$$

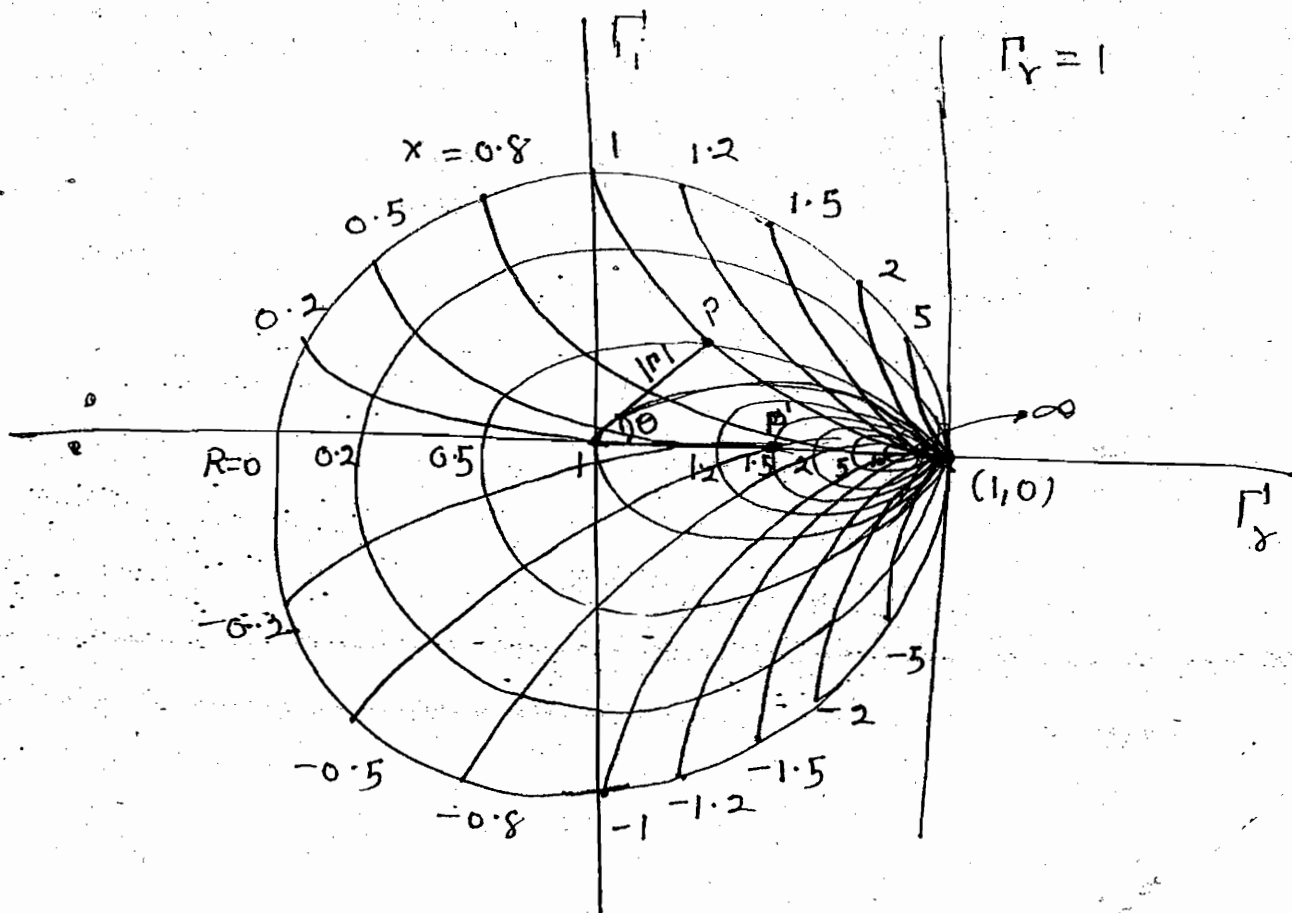
Constant R circles
Equation.

$$(\Gamma_Y - 1)^2 + \left(\Gamma_I - \frac{1}{X} \right)^2 = \left(\frac{1}{X} \right)^2$$

constant X circles
Equation.

Properties of constant R circles:-

- (a) Centres - $\left(\frac{R}{R+1}, 0 \right)$
- (b) Radius - $\left(\frac{1}{R+1} \right)$
- (c) They all pass through $(1, 0)$
- (d) Range $(0, \infty)$



1. The $R=0$ circle is the biggest possible R circle and it is periphery or boundary of Smith Chart.
2. All circles $R=0$ to 1 extend into the four quadrants the circles with $R>1$ confined in the Ist and IV quadrants
3. All the R -circles are concurrent with their centres on a straight line and have a common tangent $\Gamma_r = 1$ axis

Properties of constant X-circles:-

- a) Centres - $(1, \frac{1}{X})$ (b) Radius - $(\frac{1}{X})$
 - c) They all pass through $(1, 0)$
 - d) Range $(-\infty, \infty)$
1. $x=0$ circle is the biggest possible X circle and it is Γ_r axis
 2. All circles with $x=0$ to 1 extend into the first two quadrant but circles with $x>1$ confined in the 1st quadrant
 3. All the x -circles are concurrent and have a common tangent with Γ_r axis

Note:-

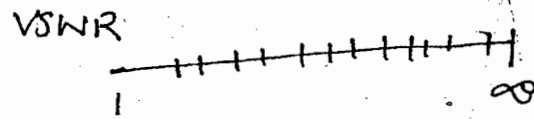
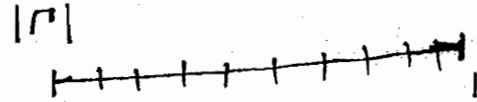
1. The centres of R circles lie on the common tangent of X -circle and vice-versa. Hence R and X constitute a pair of orthogonal families of circles.
2. A horizontal movement on the $x=0$ circle as R value increasing and a clockwise movement on the $R=0$ circle as x increasing inductively (trly)

Calculation of Γ :-

For a known Z_L and Z_0 calculate

$\frac{Z_L}{Z_0} = R + jX$ and locate the corresponding point of intersection of R and X circle

- Join the point P to origin the OP length = $|\Gamma|$
- OP segments inclination with Γ_r axis is Γ 's phase
- End



Calculation of VSWR :-

- For a known $|\Gamma|$ identify a point P' on the the $|\Gamma|$ real axis such that OP' length = $|\Gamma|$. The R circle value at P' is VSWR

$$\left. \begin{array}{l} R + jX \\ X = 0 \\ \Gamma_r \text{ axis} \end{array} \right| = \frac{Z_L}{Z_0} = \text{VSWR}$$

Identify following points on the Smith's Chart :-

(I) $Z_L = jR_0$, $Z_0 = R_0$

$$\frac{Z_L}{Z_0} = 0 + j1$$

$\downarrow$ $\downarrow$
R=0 X=1

$$\Gamma = j$$

(II) $Z_L = -jR_0$

$$Z_0 = R_0$$

bottom

(III)

$$Z_L = 0$$

$$Z_0 = R_0$$

$$\frac{Z_L}{Z_0} = 0 + j0$$

↓ ↓
R X

$$\Gamma = -1$$

Extremely left

(IV)

$$Z_L = \infty$$

$$Z_0 = R_0$$

Extremely right

(V)

$$Z_L = R_0$$

$$Z_0 = R_0$$

→ centre $\Gamma = 0$

Lecture -14

Impedance Matching Technique:-

Active
(Bias)
CB Amp
CC Amp

Passive
Series $\lambda/4$
↓
Quarter wave transformer
Shunt
↓
Stub matching

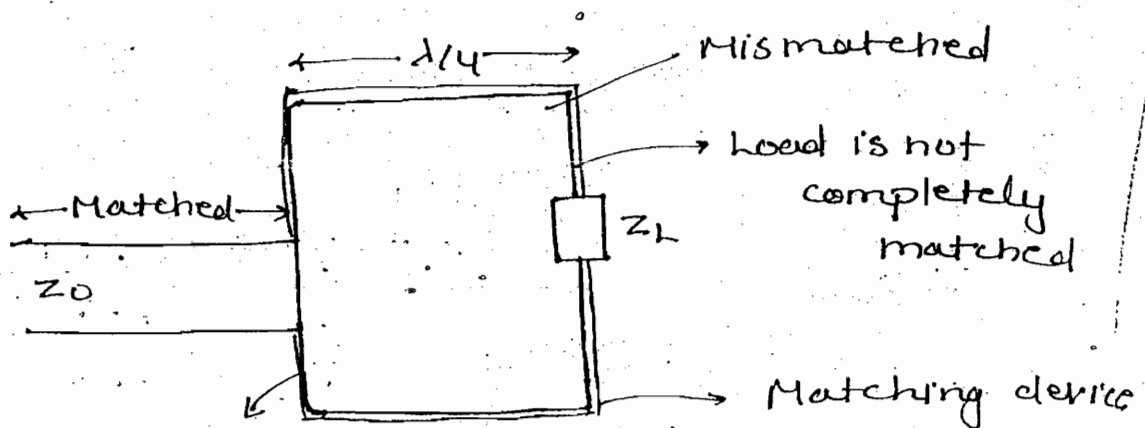
$$Z_0 = \frac{60 \ln(b/a)}{\sqrt{\epsilon_r}}$$

= few 10Ω to 100Ω

$$Z_L = \text{few } \Omega - k\Omega$$

($Z_L \neq Z_0$)
Most often)

Series ($\lambda/4$) Quarterwave transformer:-



The line is matched

The termination of Z_0 line = Z_{in} of $\lambda/4$ matching device

$$= \frac{Z_0^2}{Z_L} = Z_0$$

Design needs $\rightarrow \boxed{Z_0' = \sqrt{Z_0 Z_L}}$

eg:- $50\Omega \rightarrow \text{line} \quad 100\Omega \rightarrow \text{load}$

$$\Gamma_{(load)} = \frac{100-50}{100+50} = \frac{1}{3}$$

Using a $\lambda/4$ transformer

$$Z_0' = \sqrt{50 \times 100} = 70\Omega$$

$$\Gamma_{(at load)} = \frac{100-70}{100+70} < \frac{1}{3}$$

Disadvantage:-

i) The technique is suited for a narrow range of frequencies as the length of the transformer is frequency dependent.

ii) $Z_0' = \sqrt{Z_0 \cdot Z_L} \rightarrow$ The technique is suited only for lossless line and resistive load.

Shunt - Stub matching:-

A stub is S.C or O.C line of a pre-calculated length, placed at a precalculated position such that the line is matched from the stub to the source.

Design of a stub:-

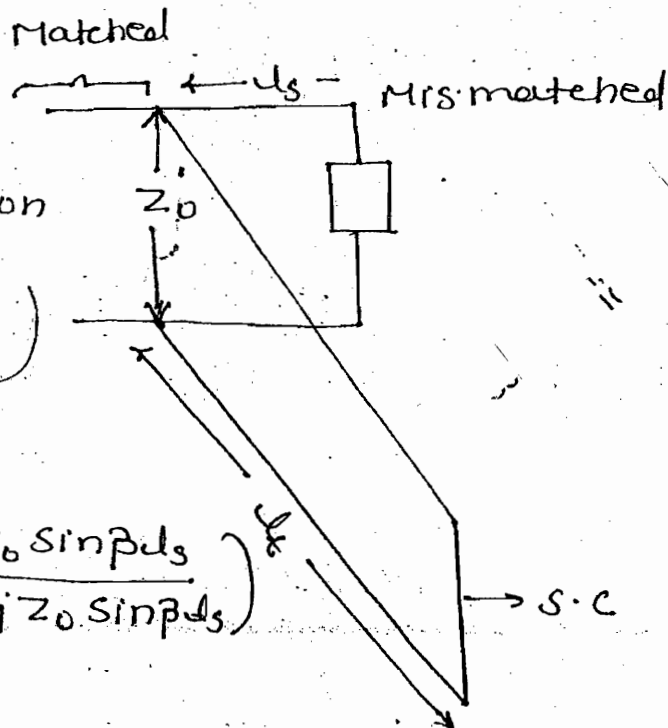
Step-(1):-

Identify a length l_s on the line from the load where the impedance has a real part Z_0 .

$$Z(l_s) = Z_0 \pm jX$$

l_s = stub position

$$l_s = \frac{d}{2\pi} \tan^{-1} \left(\sqrt{\frac{Z_L}{Z_0}} \right)$$



$$Z(l_s) = Z_0 \left(\frac{Z_L \cos \beta l_s + j Z_0 \sin \beta l_s}{Z_0 \cos \beta l_s + j Z_L \sin \beta l_s} \right)$$

$$Z(l_s) = \text{Real} \left[\frac{Z_L \cos \beta l_s + j Z_0 \sin \beta l_s}{Z_0 \cos \beta l_s + j Z_L \sin \beta l_s} \right] = 1$$

Step-(II):-

At this identical position place an equal and opposite reactance in shunt such that cancel to the existing reactance i.e.,

$$Z(l_s) = Z_0 \pm jX \quad \text{---} \quad \text{Stub}$$

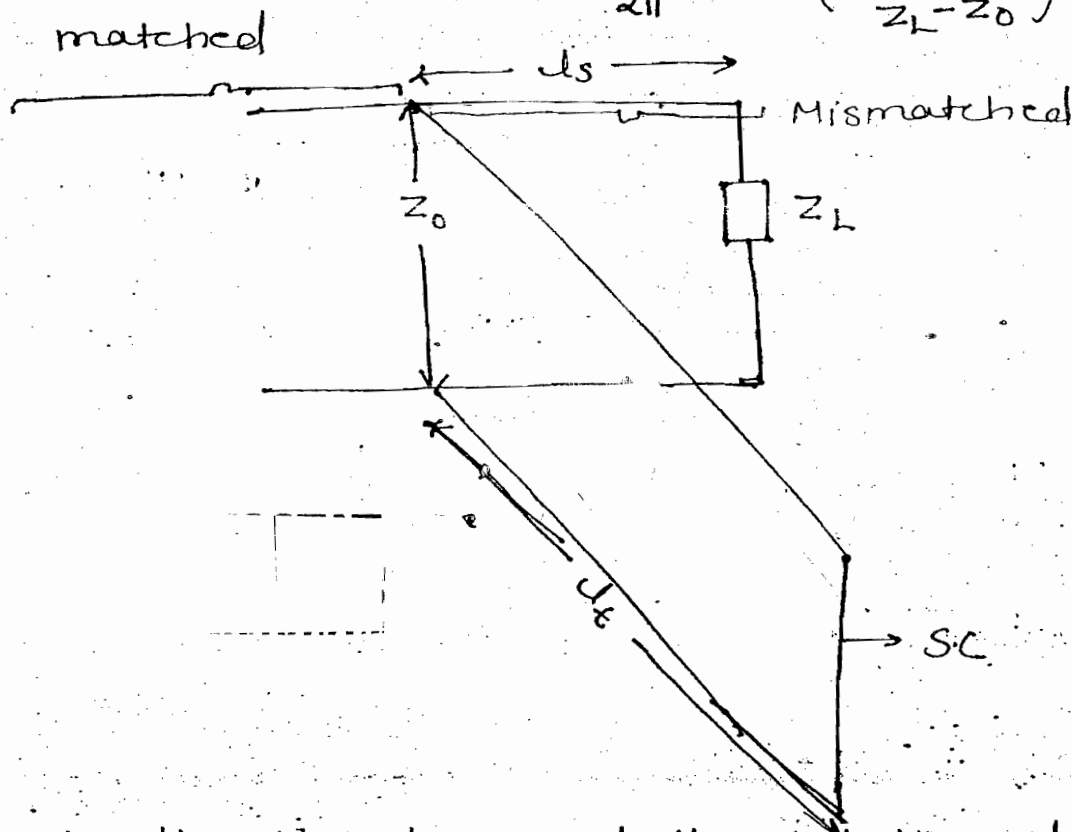
$$Y(l_s) = Y_0 \pm jB \quad \text{---} \quad \text{Stub}$$

Step-(III):-

This reactance is realised from a SC or OC stub of a finite length l_t such that

$$Z_{sc} = jZ_0 \tan \beta l_t = -\text{Imag}[Z(l_s)]$$

$$l_t = \text{stub length} = \frac{\lambda}{2\pi} \tan^{-1} \left(\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0} \right)$$



The junction impedance at the stub and the line purely real as Z_0 and hence matched from the stub to the source.

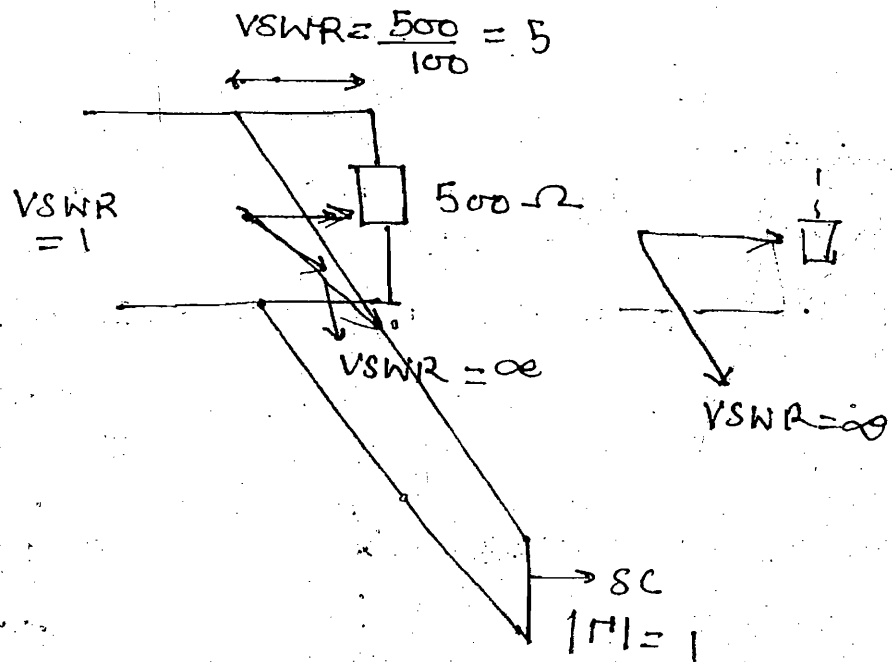
As the frequency on the line changes the stub length and stub position both have to be adjusted which is mechanically inconvenient.

however in SC stub it is easy to using movable shorts

NOTE:-

For wide range of frequencies we use double stub matching where l_{s1} and l_{s2} are two stub position fixed and l_{t1} and l_{t2} are varied.

WORKBOOK:-



$$VSWR = 3$$

$$|\Gamma|_V = \frac{3-1}{3+1} = \frac{1}{2}$$

$$\Gamma_P = -\frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4} = 25^\circ$$

$$Z_o' = \sqrt{225 \times 250} = 240 \Omega$$

$$Z_L = 75 - j40$$

$$Z_o = 75$$

Note:-

Stub can never be placed at node

OC $\rightarrow$ Stubs $\rightarrow$ More difficult to realize

SC $\rightarrow$ Stubs

Ans-(a)

36.

The diagram shows a horizontal cross-section of a co-axial cable. The top half is labeled "co-axial cable" and contains a dielectric material with impedance Z_0' . The bottom half is labeled "Air filled" and contains air with impedance Z_0 . The cable is divided into three sections by vertical boundaries. The left section is labeled $50-\Omega$, the middle section is labeled $72-\Omega$, and the right section is labeled $72-\Omega$. The length of the middle section is indicated by a double-headed arrow and labeled 4λ .

$$\ln(b/a) = 1 \Rightarrow \frac{b}{a} = e = 2.71 \Rightarrow b = 27 \text{ mm}$$

37. $0.5 - j0.3 = \frac{Z_L}{Z_0} = \text{Normalized value}$

$$\begin{aligned}\text{Actual value} &= (0.5 - j0.3) \times 50 \\ &= 25 - j15\end{aligned}$$

38. clocking on
 $R = \text{constant}$

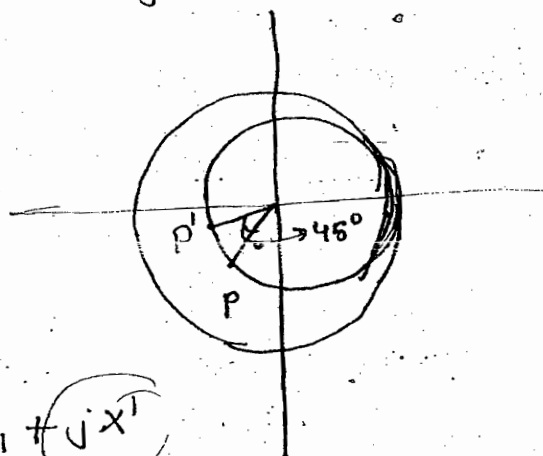
$$Z \text{ at } P = R + jX_L$$

$$Z' \text{ at } P' = R + jX_2$$

with $x_2 > x_1$

$$Z' \text{ at } P' = R + jX_1 + jX' \quad \text{inductive}$$

Ans - (c)



39. (1) $\frac{50 + j50}{50} = \underset{\substack{\downarrow \\ R}}{1} + \underset{\substack{\downarrow \\ X}}{j1}$

$$(11) \quad \frac{10 + j10}{50} = 0.2 + j0.2$$

$$(iii) \quad \frac{25 - j50}{50} = 0.5 - j1$$

40 (1) one revolution for 360° - length - $1/2$
 $\downarrow$
 All impedances repeat for $1/2$

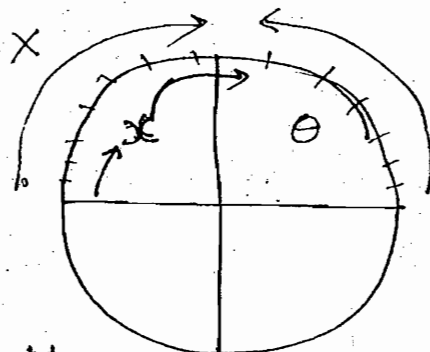
- (II) Clockwise movement $X \uparrow$
towards generator $x \uparrow$

$$Z(x) = Z_0 \left[\frac{Z_L \cos \beta x + j Z_0 \sin \beta x}{Z_0 \cos \beta x + j Z_L \sin \beta x} \right]$$

$$= j Z_0 \tan \beta x$$

$$x \uparrow \quad X \uparrow$$

(III) $\Gamma(x) = \Gamma(0) e^{-j2\beta x}$



Three scales

→ X

→ θ

→ x

Three scales on the Boundary

- (I) X - Reactance of the load
- (II) x - Length on the line
- (III) Γ 's phase

All the three are related scales

→ one is sufficient

(IV) Impedance (Wrong Statement)

Conventional :-

$$V(x) = \frac{V_L (Z_L + Z_0)}{2Z_L} \left[e^{r x} + \frac{Z_L - Z_0}{Z_L + Z_0} e^{-r x} \right]$$

$$= V_L \cosh r x$$

$$\text{If } r = j\beta \quad V(x) = V_L \cos \beta x \quad \text{--- (1)}$$

OR

$$V(x) = V_L \cosh r x + I_L Z_0 \sinh r x$$

$$V(x) = V_L \cosh r x$$

$$I(x) = I_L \cosh(r x) + \frac{V_L}{Z_0} \sinh(r x)$$

$$I(x) = \frac{V_L}{Z_0} \sinh(\gamma x) \quad \text{and} \quad \text{voltage}$$

$$\text{If } \gamma = j\beta \quad I(x) = \frac{jV_L}{Z_0} \sin \beta x$$

$$V_L = 12V$$

$$Z_0 = 60 \Omega$$

$$x = \lambda/4$$

$$d = \lambda/2$$

$$\left. \begin{array}{l} V_L = 12V \\ Z_0 = 60 \Omega \\ x = \lambda/4 \\ d = \lambda/2 \end{array} \right\} \rightarrow \text{if } \gamma = j\beta \quad V(x) = V_L \cos \beta x$$

$$Z_L = 100 - \frac{j}{\omega C} = 100 - j180 \Omega$$

$$= 100 - \frac{j}{2 \times \pi \times 10^8 \times \frac{1}{36 \pi \times 10^9}}$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \quad \text{VSWR}_0$$

$$Z_{\max} = Z_0 (\text{VSWR})$$

$$Z_{\min} = Z_0 / \text{VSWR}$$